Week 1 & 2 Worksheet

Worksheet Details:

You will use this worksheet to record your responses to all activities that require you to do so in the instructions. Each time an activity references this worksheet to record your responses, you must open this copy of your worksheet and add your response in the appropriate section. This specific worksheet will be **used for all Weeks 1 and 2 activities**.

To complete the assignment, you must follow these steps:

1. Complete this worksheet with the responses from your activity.
2. You'll find that each question has its own submission block where you must put your answers.
3. You must complete all the blocks available under a question. Leaving a block empty will mean the question is not being answered (even if you answer it somewhere else).
4. Make sure you place the correct answer under the right question. Misplaced answers will not be marked.
5. Save and submit this worksheet to Savanna.

Instructions:

1. You must fill out this worksheet and submit it as your assignment response.
2. Use tools like Foxit, Adobe Acrobat, or PDFGear to add your responses in the blank **spaces provided under a question or set of instructions**.

3. Read the instructions and questions carefully before responding.
4. **Do not** try to edit any part of this worksheet!
5. Save this worksheet using the convention "*FirstName_LastName_Assignment04_Week1&2.pdf*".

Week 1 - Activity A: Introduction to Software Engineering

Go through the instructions provided on Savanna and complete the following tasks.

Task 0: Foundation

In this task, you will complete a fill-in-the-blank activity covering key software development aspects. This will help solidify your understanding of basic computing concepts and how they fit into the bigger picture of building software.

To demonstrate your grasp of these essential concepts, fill in the blanks using the best options from the list provided.

Instructions:
Go through the fill-in-the-blank statements below, and find the appropriate terms or phrases from the list provided that fit each blank the best. Provide your answers in the space provided (the fill-in-the-blank spaces). Make sure to match the correct answer to the appropriate blank using the numbers provided. Incorrectly matched responses will be marked wrong.

Text:
In software development, the process starts with understanding the problem **(1)** the software aims to solve. Developers gather require **(2)** from users to define what the software should do. After this, they design the software's architec **(3)** to outline the system's structure. The actual coding uses prograr **(4)** languages like Python or Java. To manage changes in the code, developers use a tool called version **(5)**. Finally, before the software is deployed, it must be tested using testing **(6)** to ensure it works correctly.

Term/Phrase List:

Data	Problem	Testing	Design
Requirements	Programming	Version Control	Debugging
Security	Architecture	Codebase	Deployment
Maintenance			

Note: Fill in the blank spaces in the text above with the correct option.

Task 1: Life Cycle

You'll write down all the Software Development Life Cycle steps in this task. This will help test your understanding of the sequential steps in creating software to assess your knowledge of the fundamental processes and methodologies used in software development.

In the space provided below, list all the software development life cycle phases in order.

1.	Planning
2.	Design
3.	Development
4.	Testing
5.	Deployment

6.	Mantenance

Task 2: Explore Software Engineering Roles

In this task, you will demonstrate your understanding of the responsibilities and duties associated with each position by matching descriptions to their corresponding roles. This will help you identify potential career paths within software engineering and assess your interests and skills concerning these roles.

Match the following descriptions with their respective career roles:

- **A.** Designs, develops, tests, and maintains software applications.
- **B.** Defines the overall structure and design of software systems.
- **C.** Analyses business requirements and translates them into technical specifications.
- **D.** Writes code to implement software features and functionality.
- **E.** Tests software to ensure it meets quality standards and is free of defects.
- **F.** Manages and maintains databases used in software applications.
- **G.** Bridges the gap between software development and IT operations, focusing on automation and collaboration.
- **H.** Designs the user interface and user experience of software applications.
- **I.** Protects software systems from security threats and vulnerabilities.
- **J.** Creates documentation, manuals, and tutorials for software products.

Career Roles:

DevOps Engineer	Quality Assurance Engineer	Cloud Architect
Software Engineer	Data Scientist	Security Analyst

Software Architect	Software Developer	Database Administrator
UI/UX Designer	Technical Writer	Systems Analyst

Your Response:

In the space below, match the above descriptions with their correct career titles.

Description List	Career Title
A. Designs, develops, tests, and maintains software applications.	Software Engineer
B. Defines the overall structure and design of software systems.	Software Architect
C. Analyses business requirements and translates them into technical specifications.	Systems Analyst
D. Writes code to implement software features and functionality.	Software Developer
E. Tests software to ensure it meets quality standards and is free of defects.	Quality Assurance Engineer
F. Manages and maintains databases used in software applications.	Data Scientist

G. Bridges the gap between software development and IT operations, focusing on automation and collaboration.	DevOps Engineer
H. Designs the user interface and user experience of software applications.	UI/UX Designer
I. Protects software systems from security threats and vulnerabilities.	Security Analyst
J. Creates documentation, manuals, and tutorials for software products.	Security Analyst

Week 1 - Activity B: Introduction to Computers and How They Work

Go through the instructions provided on Savanna and complete the following tasks.

Task 0: Information Processing

In this task, you will demonstrate your understanding of the key components of a computer system and their roles in processing information by matching the definitions with the correct terms.

Task:

List the most appropriate terms that fit the descriptions below:

A. I am often referred to as the computer's "brain." I execute instructions and perform calculations.
B. I am a printed circuit board, and I connect all computer components.
C. I am an input device that converts physical documents into digital images
D. I manage the computer's resources and provide an interface for users to interact with the hardware.

Your Response:

In the space below, provide the term you think matches the above definition. Match your term to the definition; otherwise, it will be marked wrong.

A.	CPU Central Processing Unit
B.	Motherboard
C.	Scanner

D.	Operating system

Task 1: Acronyms

This task tests your knowledge of common acronyms used in computer systems. By understanding these acronyms, you can communicate more effectively with others in the technology field and better comprehend technical concepts and discussions.

Task:

Concerning Computer Systems, what do these acronyms stand for?

A. RAM
B. CPU
C. HDD
D. USB
E. HDMI
F. BIOS
G. CLI

Your Response:

In the space below, provide the term you think matches the above acronym. Match your term to the definition; otherwise, it will be marked wrong.

A.	Random Accse momory
B.	central processing unit
C.	hard drive disk

D.	universal serial bus
E.	high definition multimedia interface
F.	basic input/ output system
G.	command line interface

Task 2: Memories

This task tests your knowledge of different types of computer memory and their characteristics. By matching the descriptions to the correct types of memory, you will demonstrate your understanding of how data is stored and accessed in a computer system.

Task:

State the types of computer memory described in the statements below:

- **A.** The most common type of primary memory that allows data to be accessed in any order.
- **B.** A type of non-volatile memory that stores data permanently and can only be read from.
- **C.** A magnetic storage device that uses spinning disks to store data.
- **D.** A type of non-volatile memory that uses flash memory to store data

Your Response:

In the space below, provide the type of computer memory you think matches the above description. Match your term to the definition; otherwise, it will be marked wrong.

A.	RAM-random acces memory
B.	ROM-read only memory
C.	HDD-hard drive disk
D.	SSD-solid state drive

Task 3: The Roles of Operating Systems

This task tests your understanding of the functions of an operating system. By identifying the incorrect options, you demonstrate your knowledge of the roles and responsibilities of an operating system in a computer system.

Task:

Among the following, select the **3 INCORRECT FUNCTIONS** of an operating system:

A. Web browser development
B. File system management
C. Data Storage
D. Hardware control
E. Application development
F. Process management
G. Network management

Your Response:

In the space below, provide the list of functions (from among the ones provided above) that are **NOT** functions of an operating system.

web browse development
application development
data storage

Task 4: Practical Application

This task tests your knowledge of popular operating systems and their key characteristics. You will demonstrate your understanding of the operating system landscape and its common uses by correctly identifying the operating systems that match the given descriptions.

Task:

Mention the operating system commonly used for x, y, and z:

X - has a sleek design, focuses on user experience, and is integrated with Apple devices.
Y - Developed by Microsoft and is the most widely used desktop and laptop OS.
Z - A family of open-source operating systems known for their flexibility and customization.

Your Response:

In the space below, provide the operating system commonly used for X, Y, and Z, as described above. Match your term to the definition; otherwise, it will be marked wrong.

X.	Mac OS
Y.	windows
Z.	Kali linux

Week 2 - Activity A: Version Control Systems

Go through the instructions provided on Savanna and complete the following tasks.

Task 0: Functions of Version Control Systems

This question tests your knowledge of Version Control Systems (VCS) and their functions. By identifying the incorrect functions, you will demonstrate your understanding of the key roles and responsibilities of a VCS in software development.

Task:

Among the following, select the **5 INCORRECT** functions of a version control system.

- **A.** Tracking changes to files
- **B.** Creating documentation
- **C.** Reversing changes to previous versions
- **D.** Compiling code
- **E.** Resolving conflicts between different versions
- **F.** Creating backups of files
- **G.** Running automated tests
- **H.** Generating reports on code changes
- **I.** Formatting code
- **J.** Collaborating with multiple developers
- **K.** Debugging code

Your Response:

In the space below, write out the items from the list above that are **NOT** functions of a version control system.

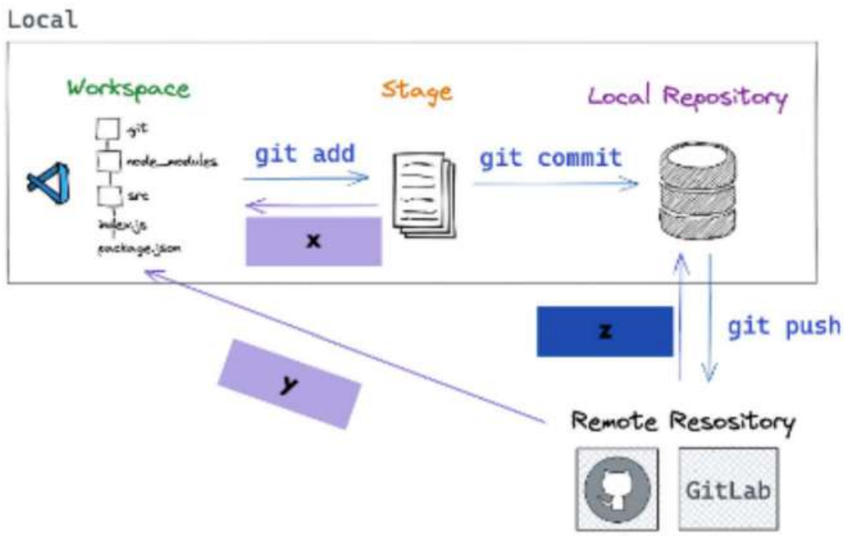
creating documentation
compiling code
creating backups of files
formating code
debugging code

Task 1: Git Commands

This task tests your knowledge of Git commands and their corresponding functions. By correctly identifying the Git commands for the given processes, you will demonstrate your understanding of Git's core features and how they are used in version control.

Task:

Name the git commands used for processes X, Y, and Z in the below process diagram.



Your Response:

In the space below, write the commands for the processes X, Y, and Z shown above. Make sure to match your commands to the process; otherwise, they will be marked wrong.

X.	git clone
Y.	git clone
Z.	git clone

Task 2: GitHub Essentials

This task tests your knowledge of Version Control Systems (VCS) and their functions. By correctly identifying the references that apply to VCSs, you will demonstrate your understanding of the key roles and responsibilities of a VCS in software development.

Task:

Select the **6 CORRECT** references of Version Control Systems from the list below.

- **A.** A system that tracks changes to files over time.
- **B.** A type of programming language.
- **C.** A tool for managing multiple versions of a project.
- **D.** A software testing tool.
- **E.** A web browser.
- **F.** A collaborative software development tool.
- **G.** A hardware component of a computer.
- **H.** A cloud storage service
- **I.** A method for tracking and controlling changes to source code.
- **J.** A presentation software.
- **K.** A way to undo changes and revert to previous versions.
- **L.** A tool for preventing conflicts and ensuring data integrity in software development.

Your Response:

In the space below, write out the items from the list above that reference Version Control Systems.

A system that tracks changes to files over time.
A tool for managing multiple versions of a project.
A collaborative software development tool
A method for tracking and controlling changes to source code.
A way to undo changes and revert to previous versions.
A tool for preventing conflicts and ensuring data integrity in software development.

Task 3: Best Practices

This task tests your knowledge of best practices for using Version Control Systems (VCS). By listing the best practices, you will demonstrate your understanding of effectively managing software project changes, collaborating with others, and maintaining a clean and organized codebase.

Task:

Select the **3 BEST** practices for using Version Control Systems from the options below.

- Commit frequently
- Ignore conflicts
- Rebase frequently

- Use branches effectively
- Write descriptive commit messages
- Commit everything

Your Response:

Write the **3 BEST** practices for using Version Control Systems in the space below.

Use branches effectively
Write descriptive commit messages
commit frequently

Week 2 - Activity B: Introduction to Scratch Programming

Go through the instructions provided on Savanna and complete the following tasks.

Task 0: Programming is the Language of the Future

This question assesses your understanding of programming and its applications. By correctly identifying the definition of programming and its role in creating software, apps, and games, you will demonstrate your knowledge of basic programming concepts and their impact on our daily lives.

Task:
Among the following, select **THE BEST** description of "programming."

- **A.** Programming is the art of teaching computers how to solve problems. It's like giving directions to a friend who doesn't know the area. By providing computers with clear instructions, programming enables us to create software, apps, and games.
- **B.** Programming is the process of designing and building hardware components. It's like assembling a piece of furniture from instructions. Programming allows us to create physical devices like smartphones and laptops.
- **C.** Programming is the process of creating instructions for a computer to follow. It's similar to writing a recipe with precise steps and ingredients. Programming allows us to build software, apps, and games by breaking down complex tasks into smaller, manageable steps.
- **D.** Programming is the study of human behavior. It's like understanding how people think and act. Programming enables us to create software that mimics human intelligence.

Your Response:

In the space below, write out the item from the list above that best describes "programming."

Programming is the process of creating instructions for a computer to follow. It's similar to writing a recipe with precise steps and ingredients. Programming allows us to build software, apps, and games by breaking down complex tasks into smaller, manageable steps.

Task 1: Language Matters, Choose Wisely

The question's objective is to assess your knowledge of popular programming languages and their applications. By filling in the blanks with the most appropriate options, you will demonstrate your understanding of the strengths and weaknesses of different languages and your ability to select the best tool for a given task. This exercise will help you develop your programming skills and make informed decisions when working on software projects.

Task:

Fill in the blanks with the most appropriate options. **You can use an option more than once.**

As a software developer, I want to choose the most suitable programming language for my project based on its specific requirements. I need a versatile, efficient, and well-suited language for the task. For example, if I'm building a complex web application, I might consider using Kotlin___ **(1)** or Python___ **(2)** for its ease of use and robust frameworks. If I need to develop a high-performance system, C++___ **(3)** could be a good choice. And if I'm creating interactive web pages, Javascri___ **(4)** is essential. By understanding the strengths and weaknesses of different programming languages, I can make informed decisions and select the best tool for the job.

Options:

- Python
- C++
- Swift
- Go
- Kotlin
- Javascript

Note: Fill in the blank spaces in the text above with the correct option.

Task 2: Understanding Scratch is the Key to Unlocking Your Creativity

The question's objective is to test your understanding of Scratch terminology and its core components. By filling in the blanks correctly, you will demonstrate your knowledge of the key concepts and elements used in creating Scratch projects. This will help you develop your skills in using Scratch for programming and creating interactive projects.

Task:

Fill in the blanks with the most appropriate options. **You can use an option more than once.**

In Scratch, a _sprite___ **(1)** is like a character you can add to the _stage___ **(2)**, the virtual world where your project takes place. You can use costumes **(3) to give your sprite different looks**. You can also add _sounds_ **(4)** to make your project more engaging. If you want to draw or create graphics, use the _pen____ **(5)** tool. The backdrop_ **(6)** is the background of your stage. To control your sprites' actions, you'll create _scripts **7)** using _blocks **8)**. Each block represents a specific action or command; you can stack them together for your desired behavior. You can also use variables_ **(9)** to store data, lists **(10)** to store multiple values, and broadcast **(11)** to send messages between sprites.

Terms List (Options):

Backdrop	Sprite	Variables	Sounds
Event	Pen	Scripts	Loops
Blocks	Lists	Broadcasts	Costumes
Stage			

Note: Fill in the blank spaces in the text above with the correct option.

Once you have completed this worksheet:

1. Save as .pdf.
2. Rename it per the instructions.
3. Upload to Savanna as your Week 4 Assignment Submission.
4. **Celebrate a job well done!**